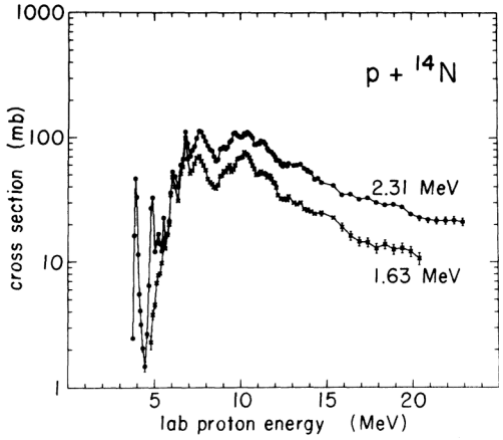
Name: _____

Due: November 13, 11:59PM via e-mail or Edwards Mailbox

1. Suppose we want to measure ^{14}N(α,n). What is the minimum α energy we would need to initiate this reaction on a ^{14}N target? What is the minimum ^{14}N energy we would need to initiate this reaction on a ^{4}He target?

2. Estimate the proton-capture cross section for ^{14}N at a for a proton lab energy of 6 MeV. Compare your result to the results of P. Dyer *et al.* PRC **23** 1865 (1981).



3. What are the natural parity states in ^{18}Ne that can be populated in ^{14}O(α,p) for $\ell = 0,1,2,\ldots$?

4. Consider the reaction ^{96}Zr(α,n)^{99}Mo at 2.5 MeV/u. Using a kinematics calculator, determine the maximum and minimum neutron energies for two cases: (a) A ^{4}He beam impinging on a ^{96}Zr target, (b) a ^{96}Zr beam impinging on a ^{4}He target. In both cases, provide a copy of the kinematics calculator plot of neutron laboratory energy vs. laboratory angle.

5. According to R. Hannaske *et al.* EPJA **49**, 137 (2013), the ^{197}Au + n cross section is ~6b at E_n = 500 keV. Estimate the neutron-capture cross section at E_n = 100 keV. Compare to their value of ~11b.

6. Consider a resonant (α, p) cross section where, for this particular resonance $\Gamma_\alpha \gg \Gamma_p$ and $\Gamma_{total} \approx \Gamma_\alpha + \Gamma_p$. Which of these three quantities has the biggest impact on the cross section for this resonance? Why?

7. What is the compound nucleus resulting from protons bombarding an aluminum target? Provide at least five different ways for the compound nucleus to decay.

8. The (d,p) reaction on ^{52}Cr leads to the $\frac{3}{2}^-$ ground state of ^{53}Cr. How would the analysis of the angular momentum transfer in this reaction differ between an analysis in terms of direct reactions and one in terms of compound-nucleus reactions?

9. Choose a reaction mechanism from the "Other Types" lecture and describe the formalism. Highlight a recent result from the literature using this reaction mechanism.

10. How would you "un-grade" yourself on this assignment? What feedback would you give yourself upon completing this homework?